# Terms of Reference (ToR)

## Clean Water Well — Nkwen-Mbatu Village

| Field | Detail |
|---|---|
| **Reference** | ADA/2024/002 |
| **Project Title** | Borehole Drilling and Solar Pump Installation — Nkwen-Mbatu |
| **Implementing Organization** | Apoti Development Association (ADA) |
| **Location** | Nkwen-Mbatu Village, ~15 km from Bamenda, North-West Region, Cameroon |
| **Duration** | June 1 – October 31, 2024 (5 months) |
| **Total Budget** | 2,700,000 CFA (~$4,500 USD) |
| **Direct Beneficiaries** | 300+ villagers |

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## 1. Background and Context

### 1.1 Community Situation

Nkwen-Mbatu is a rural village of over 300 inhabitants located approximately 15 km from Bamenda, the regional capital of the North-West Region of Cameroon. The village sits on the volcanic highlands of the Bamenda Massif, characterized by weathered basaltic and granitic formations with significant groundwater potential at moderate depths.

Prior to this intervention, the entire community relied on a single surface water source — a stream located approximately 2 km from the village center. This stream, shared with livestock and subject to agricultural runoff, was the primary source of waterborne diseases including typhoid, cholera, and dysentery that regularly affected the population.

### 1.2 Impact on Women and Girls

Women and girls bore the disproportionate burden of water collection, spending 2–4 hours daily on the round trip to the stream. Girls frequently missed school during dry season when the stream receded and collection times increased. This perpetuated a cycle of educational disadvantage and limited economic opportunity for women in the community.

### 1.3 Geological Context

The North-West Region of Cameroon sits on the Cameroon Volcanic Line (CVL), a chain of volcanic and plutonic formations extending from the Atlantic Ocean into the continental interior. The Bamenda Highlands are composed primarily of:

- **Surface layer (0–5 m):** Lateritic clay and topsoil
- **Weathered zone (5–20 m):** Decomposed basalt and granite with moderate permeability
- **Fractured basement (20–50 m):** Fractured crystalline rock with significant groundwater storage capacity

Groundwater in this region typically occurs at depths of 30–50 meters within fractured zones of the crystalline basement. Yield rates of 0.5–3.0 $m^3$/hour are common for boreholes in similar geological settings. The water table is recharged by the region's substantial annual rainfall (2,000–2,500 mm).

### 1.4 Organizational Context

This project represents ADA's largest infrastructure intervention since the organization's founding in 2021. ADA's prior work focused on direct support programs for widows, orphans, and young women. The Clean Water Well project marks a strategic expansion into community infrastructure that addresses root causes of vulnerability.

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## 2. Objectives

### 2.1 Overall Objective

To provide sustainable, clean water access to the 300+ inhabitants of Nkwen-Mbatu village through the drilling of a solar-powered borehole, eliminating dependence on a distant, polluted surface water source.

### 2.2 Specific Objectives

1. Conduct a professional geological survey to identify optimal drilling location and confirm groundwater availability
2. Drill a borehole to an estimated depth of 45 meters to reach the fractured aquifer zone
3. Install a solar-powered pumping system for sustainable, zero-cost water extraction
4. Establish and train a community water management committee for long-term maintenance
5. Reduce waterborne disease incidence in the community
6. Eliminate the time burden on women and girls for water collection

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## 3. Technical Specifications

### 3.1 Geological Survey

- **Method:** Vertical Electrical Sounding (VES) using Schlumberger array
- **Coverage:** Minimum 3 survey points within the village perimeter
- **Deliverable:** Technical report identifying optimal drilling point, estimated depth to water table, and expected yield
- **Duration:** 1–2 days

### 3.2 Borehole Drilling

- **Target depth:** 45 meters (actual depth determined by geological conditions)
- **Drilling method:** Rotary percussion (air flush) suitable for crystalline basement rock
- **Casing:** 6-inch (150 mm) PVC casing with stainless steel screen at aquifer level
- **Gravel pack:** 2–4 mm graded gravel around screen section
- **Sanitary seal:** Cement grout from surface to minimum 6 meters depth to prevent contamination
- **Development:** Air-lift development until water runs clear (minimum 4 hours)
- **Pump test:** Minimum 8-hour constant rate test to determine sustainable yield
- **Expected yield:** 0.5–2.0 $m^3$/hour
- **Water quality:** Must meet WHO drinking water guidelines (tested for bacteriological and chemical parameters)

### 3.3 Solar Pump System

- **Pump type:** Submersible solar pump rated for 40–50 m total dynamic head
- **Solar panels:** Polycrystalline panels, minimum 400W total capacity
- **Controller:** MPPT controller with dry-run protection
- **Storage:** Elevated tank (minimum 2,000 liters) for gravity-fed distribution
- **Distribution:** Minimum 2 tap stands within 200 m of residential areas
- **Design life:** 15–20 years for borehole; 10+ years for solar pump system

### 3.4 Piping and Fittings

- **Rising main:** HDPE pipe from pump to surface
- **Distribution:** PVC piping from tank to tap stands
- **Fittings:** Brass taps, concrete aprons with drainage channels at tap stands
- **Protection:** Lockable wellhead cover, fenced perimeter (minimum 10 m radius)

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## 4. Scope of Work

### Phase 1: Preparation (June 2024)

- Community mobilization and stakeholder meetings
- Formation of village water committee
- Geological survey and site selection
- Procurement of drilling contractor

### Phase 2: Drilling (July – August 2024)

- Site preparation and access
- Borehole drilling to target depth
- Casing installation and grouting
- Well development and pump testing
- Water quality testing

### Phase 3: Solar Pump Installation (September 2024)

- Solar panel mounting structure fabrication and installation
- Submersible pump installation
- Controller and electrical connections
- Elevated storage tank installation
- Piping and tap stand construction

### Phase 4: Training and Handover (October 2024)

- Water committee training (2-day program covering maintenance, hygiene promotion, financial management)
- Community hygiene awareness sessions
- Project documentation and signage
- Formal handover ceremony

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## 5. Project Team

| Role | Number | Responsibility |
|---|---|---|
| ADA Project Coordinators | 4 | Overall management, community liaison, monitoring, documentation |

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|---|---|---|
| Drilling Contractor | 1 (with crew) | Geological survey, borehole drilling, development, testing |
| Solar Pump Installer | 1 (with assistant) | Pump, panels, tank, piping installation |
| Village Water Committee | 5 members | Community mobilization, long-term management, maintenance |

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## 6. Expected Outcomes

1. **One functional borehole** with solar-powered pump providing clean water to 300+ people
2. **Reduced waterborne disease** — measurable reduction in typhoid, cholera, and dysentery cases within 6 months
3. **Time savings** — elimination of 2–4 hour daily water collection trips for women and girls
4. **Improved school attendance** — girls no longer missing school for water collection
5. **Trained water committee** — 5 community members capable of routine maintenance and financial management
6. **Sustainable system** — solar power eliminates recurring fuel/electricity costs

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## 7. Budget Summary

| Item | Amount (CFA) |
|---|---|
| Geological survey | 150,000 |
| Borehole drilling (45 m) | 1,500,000 |
| Solar pump + installation | 600,000 |
| Piping and fittings | 200,000 |
| Water committee training | 50,000 |
| Community mobilization | 50,000 |
| Transport/coordination | 80,000 |
| Documentation/signage | 30,000 |
| Contingency | 40,000 |
| **Total** | **2,700,000 CFA ($4,500 USD)** |

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## 8. Sustainability Plan

- **Financial:** Water committee collects modest user contributions (100 CFA/month per household) to build maintenance fund
- **Technical:** Committee trained in routine maintenance; ADA provides technical backstop for major repairs
- **Environmental:** Solar power ensures zero recurring energy costs; borehole designed for 15–20 year lifespan

- **Social:** Community ownership through water committee; inclusive representation (minimum 2 women on committee)

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## 9. Monitoring and Evaluation

- Monthly site visits during implementation
- Water quality testing at commissioning and every 6 months thereafter
- Beneficiary satisfaction survey at 3 months and 12 months post-completion
- Disease incidence tracking through local health center records
- School attendance records for girls (baseline vs. post-intervention)

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## 10. Risks and Mitigation

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Insufficient water yield at target depth | Low | High | Geological survey to confirm; contingency for deeper drilling |
| Equipment breakdown during drilling | Medium | Medium | Use experienced contractor with backup equipment |
| Community conflict over water access | Low | Medium | Inclusive committee formation; clear usage rules |
| Solar panel theft/damage | Low | High | Fenced perimeter; community watchman arrangement |
| Dry season yield reduction | Low | Low | Pump test during dry season conditions; adequate aquifer depth |

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